

Complete genome sequence of bacteriophage Godfather isolated from *Microbacterium foliorum*

Joshua Hutchings,¹ Lauren Bower,¹ Ethan Collum,¹ Timothy Hester,¹ Melody Hunter,¹ Chaney Kelly,¹ Luke Reynolds,¹ Cole Moore,¹ Dustin Edwards¹

AUTHOR AFFILIATION See affiliation list on p. 2.

ABSTRACT Microbacteriophage Godfather was collected from a soil sample in Stephenville, Texas. The 17,452-bp double-stranded genome contains 24 protein-coding genes. The genome shares >99% nucleotide sequence identity with cluster EE microbacteriophages Scamander, Danno, Kojax4, and Burgy.

KEYWORDS bacteriophage genetics

Investigating the genetic diversity of bacteriophages increases our understanding of their evolution and biological roles (1–3). We isolated the bacteriophage Godfather using *Microbacterium foliorum* NRRL B-24224, which was selected as the host bacteria for its lack of prophages and internal defense systems (4, 5). A soil sample was collected from a flower bed in Stephenville, Texas (GPS coordinates 32.21191 N, 98.211755 W) and incubated with peptone yeast calcium (PYCa) media at 29°C in a shaking incubator for 2 hours (6). The sample was centrifuged at 10,000 × *g* for 20 minutes, and the supernatant was collected and filtered (0.22 μL pore size), inoculated with a freshly saturated *M. foliorum* culture, and incubated at 29°C on a PYCa plate in a soft agar overlay (6). At 24 hours, Godfather formed 2 mm clear plaques, and the bacteriophage was isolated by two rounds of ten-fold serial dilutions and plating before being prepared as a high-titer lysate (6). A 300-mesh copper grid was prepared with high-titer lysate and stained with 1% uranyl acetate. Transmission electron microscopy, using an FEI Tecnai G2 Spirit BioTWIN (NL1.160G), showed that Godfather has siphovirus morphology (Fig. 1A). ImageJ v1.53m (7) was used to measure an approximate tail length of 115 nm and a capsid diameter of 50 nm (*n* = 6).

Genomic DNA was extracted from high-titer lysate by ZnCl₂ precipitation (8). A library was prepared with the NEBNext Ultra II kit and sequenced with an Illumina MiSeq instrument at the Pittsburgh Bacteriophage Institute (9) to 38× coverage from 30,304 single-end reads of 150 base read length. Raw reads were verified for accuracy using Consed v29.0 (9) and assembled as a single contig using Newbler v2.9 (10). The Godfather genome is 17,452 bp in length with a 9-nucleotide 3′ sticky overhang (5′-C CCGCCCCA-3′) and contains 68.7% G+C content, similar to host *M. foliorum* (5).

BLASTn (11) query of the sequence with the GenBank nonredundant/nucleotide (nr/nt) database returned nucleotide sequence identity (NSI) >99% with cluster EE bacteriophages Scamander (NC_054431), Danno (MT316462), Kojax4 (OR159657), and Burgy (ON755188) and 98% NSI to microbacteriophages Eightball (OK040783) and TimoTea (MK524502) (12–15). Of interest, Godfather shared 97% NSI with cluster EE bacteriophage Loca, which was previously isolated from a shopping cart handle at a nearby vendor that sold commercially available bagged soil (16). Starterator v546 (<http://phages.wustl.edu/starterator/>), Glimmer v3.02 (17), and GeneMark v2.5 (18) were used for auto-annotation, followed by manual refinement with Phamerator (19), PECAAN (<https://discover.kbrinsgd.org/>), and DNA Master v5.23.6 (<http://phagesdb.org/>)

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Address correspondence to Dustin Edwards, dcedwards@tarleton.edu.

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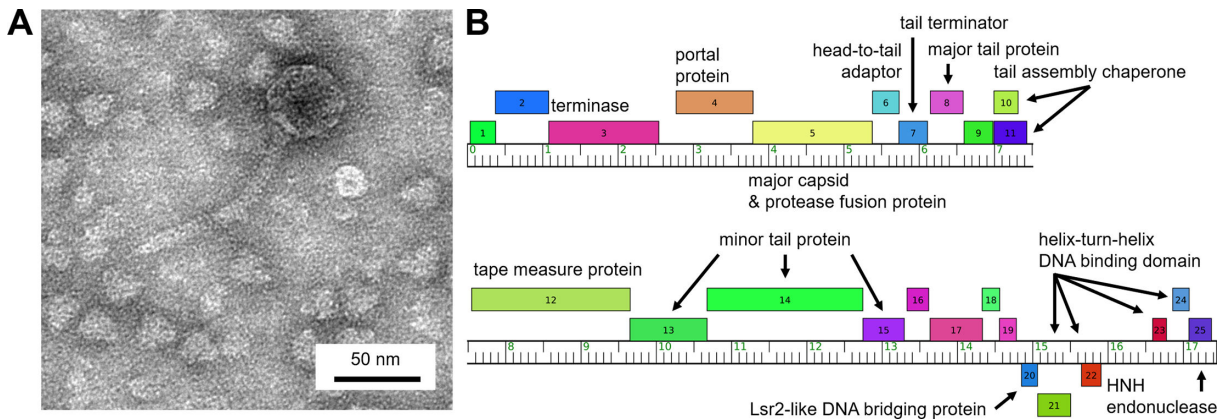


FIG 1 Bacteriophage Godfather virion and genome organization. (A) A representative micrograph of bacteriophage Godfather demonstrating siphovirus morphology with an approximate tail length of 115 nm and a capsid diameter of 50 nm. (B) Genes represented by boxes located above or below the ruler (forward or reverse orientation) and labeled with putative functions. Auto-annotation was performed by Starterator, Glimmer, and GeneMark, followed by manual refinement with Phamerator, PECAAN, and DNA Master. Protein-coding genes were identified by queries to BLASTp and HHPred.

DNAMaster/). TmHm v1.0.24 (20) identified potential transmembrane helices. Consistent with other cluster EE bacteriophages, no tRNA genes were identified using Aragorn v1.2.38 (21) and tRNAscan-SE v2.0 (22). Nineteen of 24 protein-coding genes were called using queries to BLASTp v2.14.1 (11) and HHPred v3.0beta (PDB, UniProt, Pfam-A v36, and NCBI v3.19 databases) (23). Default settings were used for all programs. Functional assignments include structural proteins, an Lsr2-like DNA-bridging protein, four helix-turn-helix DNA-binding domains, and an HNH endonuclease (Fig. 1B).

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AUTHOR AFFILIATION

¹Department of Biological Sciences, Tarleton State University, Stephenville, Texas, USA

AUTHOR ORCIDs

Dustin Edwards  <http://orcid.org/0000-0001-6409-8041>

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AUTHOR CONTRIBUTIONS

Joshua Hutchings, Formal analysis, Investigation, Writing – original draft | Lauren Bower, Formal analysis, Investigation, Writing – original draft | Ethan Collum, Formal analysis, Investigation, Writing – original draft | Timothy Hester, Formal analysis, Investigation, Writing – original draft | Melody Hunter, Formal analysis, Investigation, Writing – original draft | Chaney Kelly, Formal analysis, Investigation, Writing – original draft | Luke Reynolds, Formal analysis, Investigation, Writing – original draft | Cole Moore,

Formal analysis, Investigation, Project administration, Supervision, Writing – original draft, Writing – review and editing | Dustin Edwards, Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Visualization, Writing – original draft, Writing – review and editing

DATA AVAILABILITY

Godfather genome sequence is available at GenBank under the accession number [PP725414](#) and in the SRA under accession number [SRX24892095](#).

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